Vendor Specialization Claims at 4B: A Controlled Comparison of MedGemma and Gemma-3 on MIMIC-III MICU Nursing Tasks

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Abstract

Large language models (LLMs) marketed as pre-trained on clinical corpora are widely assumed to outperform general-purpose models on clinical natural language processing (NLP) tasks. We test this vendor claim directly for a single 4-billion-parameter checkpoint pair, comparing MedGemma-4B-IT (Google's clinically pre-trained model) against Gemma-3-4B-IT, a general instruction-tuned model of identical size and architecture, on three MICU nursing tasks drawn from MIMIC-III: structured nursing note generation (Sub-task 1), abnormal vital sign identification (Sub-task 2), and admission summary generation (Sub-task 3). Using one record per unique hospital admission ($n = 637$ independent records) and 13 evaluation metrics, the clinically marketed checkpoint provides no consistent advantage. Gemma-3-4B-IT matches or outperforms MedGemma-4B-IT on the majority of metrics, including abnormal value recall ($\Delta = -0.119$, $p < 0.001$), AARP document structure ($\Delta = -0.024$, $p < 0.001$), and named mention of out-of-range metrics in free-form summaries ($\Delta = -0.079$, $p < 0.001$). We caution that the two checkpoints differ in alignment recipe as well as pre-training corpus, so these results falsify the vendor's specialization claim for this checkpoint pair without identifying the pre-training corpus as the causal driver. We report the result as null-to-negative and hypothesise that, at 4B parameters, instruction-following quality dominates domain vocabulary as a determinant of structured clinical generation performance.

Keywords

clinical NLP, large language models, nursing documentation, MIMIC-III, domain pre-training, benchmarking

1 Introduction

The deployment of artificial intelligence in clinical settings has been accompanied by a pervasive marketing claim: that models pre-trained on medical literature, clinical notes, and biomedical ontologies will outperform general-purpose models on clinical language tasks. This framing is intuitively appealing, since a clinician trained on medical texts should know more medicine than a generalist, and it has driven considerable research effort in domain-specific pre-training pipelines for healthcare NLP [6, 9].

The practical consequence is that "clinical pre-training" on a model card risks functioning as a credibility signal that outweighs empirical evidence from the institution's own patient population and workflow when health systems evaluate LLMs for clinical decision support or automated documentation.

We argue that this signal deserves controlled scrutiny on the specific checkpoints health systems are likely to consider. Nursing documentation in the intensive care unit is a stress-test where domain knowledge should matter most: notes are highly structured (vital signs, medications, active problems, assessments), grounded in observable measurements, and subject to clinical accuracy requirements. If a vendor's clinically marketed checkpoint cannot provide a measurable advantage here, the vendor's specialization claim requires revision for that checkpoint.

This paper provides exactly that test for one checkpoint pair. We compare MedGemma-4B-IT [3] (Google's 4B-parameter model marketed as pre-trained on clinical and biomedical corpora) against Gemma-3-4B-IT [2], a general instruction-tuned model of identical size and architecture, on three tasks derived from MIMIC-III [4] MICU nursing records. We emphasise that the two checkpoints differ not only in pre-training corpus but also in alignment recipe, instruction-tuning data, and possibly base-model snapshot; our design therefore falsifies the vendor's specialization claim for this pair without identifying the pre-training corpus as the sole causal driver. To eliminate within-admission correlation, we sample one record per unique hospital admission ($n = 637$ independent admission records):

(1) **Sub-task 1**: Generate a structured MICU nursing note in Assessment/Action/Response/Plan (AARP) format from 12 hours of structured EHR context.
(2) **Sub-task 2**: Identify abnormal vital signs and laboratory values from recent observations.
(3) **Sub-task 3**: Generate a free-form prose admission summary comparable to a hospital course narrative.

Our results are unambiguous: MedGemma-4B-IT does not consistently outperform Gemma-3-4B-IT. After reclassifying near-floor metrics with trivial absolute differences as draws (Section 4), the general model is the better performer on the majority of significant metrics. We report this as an explicit null-to-negative result for this checkpoint pair and argue it has direct implications for how healthcare organisations interpret vendor specialization claims when evaluating 4B-parameter LLMs for clinical NLP applications.

2 Related Work

Clinical pre-training benchmarks. Lee et al. [6] demonstrated that BERT models pre-trained on PubMed abstracts and clinical notes (BioBERT) improved named entity recognition and relation extraction on biomedical benchmark datasets. However, these gains were measured on tasks where training and test distributions closely matched the pre-training corpus. Subsequent work by Alsentzer et al. [1] found that ClinicalBERT, pre-trained exclusively on MIMIC-III clinical notes, improved over BioBERT on NER and natural language inference tasks, yet underperformed on de-identification

benchmarks where the test-set PHI format differed systematically from the MIMIC pre-training distribution. This result illustrates a broader pattern: domain pre-training benefits are sensitive to corpus–task distribution alignment, not domain membership alone. Our study extends this line of inquiry to the instruction-following era, where pre-training data composition interacts with instruction-tuning alignment in ways that are not yet well characterised.

LLM evaluation on medical question-answering. Singhal et al. [9] evaluated PaLM and Med-PaLM on medical licensing examination questions (MedQA, MedMCQA, PubMedQA) and found that domain-specific fine-tuning substantially improved performance on knowledge-recall benchmarks. A critical distinction, however, is that multiple-choice medical QA measures knowledge retrieval, not clinical language *generation*. Generating a coherent, grounded, structured nursing note from a patient's vital signs requires capabilities (structured output compliance, numerical grounding, instruction following) that are distinct from encyclopaedic medical knowledge. Our evaluation targets this generation regime rather than the QA regime.

Automated nursing documentation. Rajkomar et al. [8] showed that EHR-grounded deep learning models could predict clinical events from structured data with high accuracy, but did not address free-text generation. More recently, Nori et al. [7] demonstrated that GPT-4 achieves expert-level performance on medical licensing examinations and clinical vignette question-answering, often matching or exceeding domain-specialised models, suggesting that strong general instruction-following can be competitive with domain fine-tuning even on clinical tasks. Our study provides a direct controlled comparison at the 4B parameter scale, where computational resource constraints make the choice between a domain model and a general model practically meaningful for resource-limited healthcare settings.

3 Methodology

3.1 Dataset

We use MIMIC-III [4], a de-identified EHR database of 58,976 hospital admissions at Beth Israel Deaconess Medical Center. We restrict to the Medical Intensive Care Unit (MICU) and filter `NOTEEVENTS` for nursing progress notes (`category = 'Nursing'`). After ETL processing (see below), the dataset comprises 103,442 records across three splits (train 71,885 / validation 16,169 / test 15,388). All experiments use the validation split, sampling one record per unique hospital admission (n = 637 independent records).

3.2 ETL Pipeline

Raw MIMIC-III CSVs are processed using Polars (lazy evaluation, chunk-safe for large joins). ITEMID harmonisation uses the MIMIC-Extract variable map [10], which maps each measurement identifier to a canonical variable name; only entries with verified status are retained. Physiological plausibility filtering applies per-variable outlier bounds from MIMIC-Extract [10]; observations outside these bounds are flagged as out-of-range. For each nursing note, we construct a *FHIR-lite* record containing:

- **Observations**: all vital sign readings from the 12 hours preceding the note's `charttime`, with out-of-range flags derived from the MIMIC-Extract physiological bounds [10], capped at 1,100 entries at prompt-construction time.
- **Labs**: all laboratory values from the same 12-hour window, capped at 150 entries at prompt-construction time.
- **Medications**: all CareVue medication administration events from the same window, capped at 250 entries at prompt-construction time.
- **Active problems**: ICD-9 diagnoses linked to the admission.
- **Hospital course**: The "Brief Hospital Course" or "Hospital Course" section extracted from the admission's discharge summary via regex, used as the reference for Sub-task 3.

Records are retained only if the density gate |obs| + |labs| + |meds| ≥ 5 is satisfied. Temporal data uses ISO 8601 relative offsets (e.g., `T-10h`) to mask MIMIC date-shifting artefacts. The per-section caps (1,100 observations, 150 labs, 250 medications) were set to keep prompts within the model context window; 99.1% of admissions fall entirely within these limits, and both models received identically constructed inputs.

3.3 Models and Inference

Both models are served via vLLM [5] on a single NVIDIA RTX 3090 (24 GB VRAM), with `tensor_parallel_size=1`. Sampling parameters are identical across all runs:

- Temperature: 0.30; top-p: 0.90; repetition penalty: 1.15.
- Max tokens: 1,024 (Sub-tasks 1 and 3); 512 (Sub-task 2).

The repetition penalty of 1.15 was necessary for MedGemma-4B-IT to suppress degenerate looping: in an initial diagnostic run on the first 500 records (predating the final n=637 corpus) without this penalty, 95% of MedGemma-4B-IT outputs were hard-truncated at the token limit with no structured content. Gemma-3-4B-IT exhibited this behaviour in only 14 of those 500 records. We applied the same penalty to both models for comparability, but did not sweep penalties per model; we therefore cannot distinguish MedGemma-4B-IT's performance ceiling at its optimal configuration from its performance under the only configuration we found that produced any output (see Section 5.3).

Sub-task 1 prompt. The system prompt instructs the model to act as an ICU nurse completing a shift note. The user turn provides all FHIR-lite fields as structured text and requests one complete AARP block per active problem, preceded by a single de-identified few-shot example drawn from the validation split. The same exemplar is used across all 637 prompts and is verified excluded from the n=637 evaluation set by `row_id`. The ground truth is the original nursing note from `NOTEEVENTS`. Full prompt text is given in Appendix A.

Sub-task 2 prompt. The model receives only the most recent reading per vital sign metric and is asked to list every abnormal value with its clinical significance. Ground truth is derived by applying standard ICU clinical reference ranges (e.g., Heart Rate 60–100 bpm, $SpO_2 \geq 95\%$) that are fixed in the inference engine; these are clinically motivated normal ranges, distinct from the statistical outlier bounds used in the ETL physiological plausibility filter. The full reference range table and prompt are given in Appendix A.

Sub-task 3 prompt. The system prompt adopts a physician persona and requests a free-form prose Hospital Course narrative. The

user turn provides all FHIR-lite fields, identical in structure to Sub-task 1. The reference is the extracted hospital course section. Full prompt text is given in Appendix A.

3.4 Evaluation Metrics

Sub-task 1.

- **ROUGE-L** (F-measure, stemmed) against the original nursing note.
- **Clinical Entity F1** (aggregate, drug surface-form, vital surface-form) via `en_core_sci_sm` scispaCy NER.
- **AARP Section Score**: proportion of active problems receiving a complete Assessment/Action/Response/Plan block within a 2400-character sliding window.

Sub-task 2. Binary precision, recall, and F1 over abnormal vital sign identification, where a value is "identified" if its label appears in the generated text.

Sub-task 3.

- **ROUGE-L** against a *relevance-filtered* hospital course (sentences whose key terms appear in `source_facts` only).
- **BERTScore F1** (`distilbert-base-uncased`) against the same filtered reference.
- **Problem Coverage**: fraction of ICD-9 problem descriptions appearing as verbatim lowercased substrings in the generated text.
- **Out-of-Range Term Mention Rate** (formerly "Key Lab Coverage"): fraction of out-of-range vital/lab *labels* (e.g., "heart rate", "lactate") present as lowercased substrings anywhere in the generated text. The metric does not check that the corresponding numeric value is reproduced or that the model identifies the value as abnormal; it measures named mention only.
- **Medication Name Mention Rate** (formerly "Medication Accuracy"): fraction of administered medications whose first whitespace token (e.g., "morphine" from "Morphine 2 mg administered via IV") appears as a lowercased substring in the generated text. The metric does not check dose, route, or that the drug is described as administered.

We use the term *mention rate* rather than "coverage" or "accuracy" to reflect that these metrics reward vocabulary overlap, not clinical grounding or numerical fidelity.

All continuous metric comparisons use the paired Wilcoxon signed-rank test (two-sided) with rank-biserial correlation r as a measure of direction consistency across paired comparisons.[1] Significance threshold: $\alpha = 0.05$. We do not apply a multiple-comparisons correction; results with marginal absolute deltas (Section 4) should therefore be interpreted with caution.

3.5 Pipeline Overview

Figure 1 illustrates the end-to-end workflow from raw MIMIC-III CSVs through evaluation.

[1]Rank-biserial r reports the proportion of paired comparisons in which one model dominates the other; it is a measure of *direction consistency*, not effect magnitude. Values near 1 indicate one model wins on nearly every pair, even when the absolute mean difference is small.

Table 1: Sub-task 1: Nursing Note Generation ($n = 637$ pairs). MG = MedGemma-4B-IT; G3 = Gemma-3-4B-IT. Δ = MG minus G3. Bold = winning model per row; $* = p < 0.05$ (uncorrected). r is rank-biserial direction consistency, not effect magnitude. Rows marked "draw" have significant p but absolute Δ below practical-meaning threshold or are dominated by recogniser noise.

Metric	MG	G3	Δ	p	r	Verdict
ROUGE-L	**0.079**	0.076	+0.003	<0.001*	0.638	draw
Entity F1	**0.059**	0.054	+0.005	<0.001*	0.610	draw
Entity F1 (drugs)	0.022	**0.025**	−0.003	0.430	0.994	n.s.
Entity F1 (vitals)	0.058	**0.074**	−0.016	0.017*	0.975	G3
AARP Score	0.000	**0.025**	−0.024	<0.001*	0.999	G3†

† Penalty-configuration confound; see text.

4 Results

All three sub-tasks were evaluated over $n = 637$ independent admission records (one record per unique `hadm_id`) drawn from the MIMIC-III validation split. Both models received identical inputs; the only experimental variable is the pre-training corpus.

4.1 Sub-task 1: Structured Nursing Note Generation

Table 1 reports corpus-level means and Wilcoxon test results. ROUGE-L scores are near-floor for both models (0.079 vs. 0.076); although the difference reaches significance, the absolute delta of +0.003 is below any threshold of practical meaning and we treat this metric as a draw. Aggregate clinical entity F1 is similarly near-floor (0.059 vs. 0.054) and we likewise treat it as a draw; the recogniser (`en_core_sci_sm`) has known weak recall on ICU drug surface forms and abbreviations, and the absolute scores indicate the metric is dominated by recogniser noise rather than genuine model differences. Both metrics also share the limitation that the ground-truth nursing note contains bedside-observed content (physical examination, subjective patient reports) not present in the structured EHR context, so neither metric measures clinical accuracy directly.

The clinically meaningful differences favour Gemma-3-4B-IT: AARP Section Score ($\Delta = -0.024$, $p<0.001$) and Entity F1 (vitals) ($\Delta = -0.016$, $p=0.017$).

The AARP result is striking but must be interpreted with care: the vast majority of MedGemma-4B-IT outputs scored zero (no complete AARP block detected), compared to a non-trivial fraction of Gemma-3-4B-IT outputs achieving structure. MedGemma-4B-IT consistently failed to follow the four-header structure even after the repetition penalty was applied. We did not sweep the repetition penalty per model (Section 5.3); we therefore cannot rule out that MedGemma-4B-IT has a different inference configuration under which it produces AARP-structured output. The result establishes that MedGemma-4B-IT fails AARP compliance *at the only configuration we found that produced any non-truncated output for it*, not that it is incapable of AARP compliance in principle.

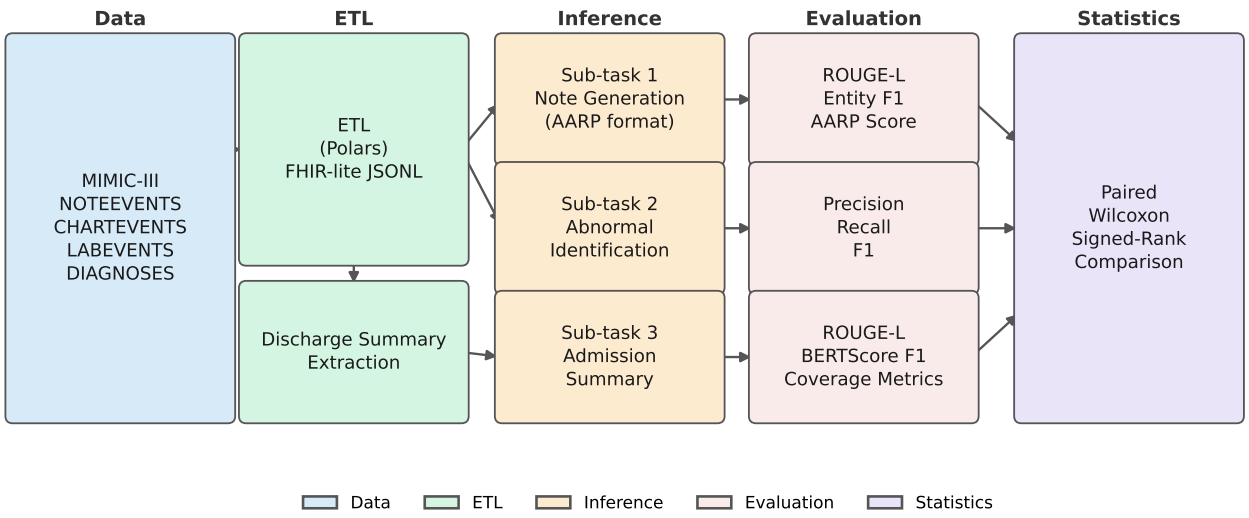

Figure 1: End-to-end pipeline. MIMIC-III raw CSVs are processed via a Polars-based ETL into FHIR-lite records. Phase 2 serves both models via vLLM on a single GPU. Phase 3 computes task-specific metrics and performs paired statistical comparison.

4.2 Sub-task 2: Abnormal Value Identification

Table 2 reports precision, recall, and F1 for identifying abnormal vital signs. All three metrics are significant. Gemma-3-4B-IT substantially outperforms MedGemma-4B-IT on recall ($\Delta = -0.119$, $p<0.001$), F1 ($\Delta = -0.056$, $p<0.001$), and precision ($\Delta = -0.029$, $p<0.001$).

The Sub-task 2 metric counts a value as "identified" if its label appears in the generated text; it does not require the model to explicitly flag the value as high or low. A more verbose model that names more vital signs in a preamble can therefore inflate its recall without performing the abnormal- flagging task. We cannot rule out that part of Gemma-3-4B-IT's recall advantage reflects greater verbosity rather than superior abnormal-value identification; a stricter re-scoring requiring explicit high/low flagging is left for future work.

Figure 2 shows per-vital miss rates. MedGemma-4B-IT misses Heart Rate at 17% versus 2% for Gemma-3-4B-IT, and Respiratory Rate at 38% versus 19%. Oxygen saturation miss rates follow the same direction (31% versus 9%). Mean arterial pressure is near-equal for both models (19% MedGemma, 22% Gemma-3), possibly because mean arterial pressure is a derived value that may not always appear as a direct `ITEMID` in CareVue; Gemma-3-4B-IT shows no consistent advantage here.

Table 2: Sub-task 2: Abnormal Value Identification ($n = 637$ pairs). Bold = winning model per row; $* = p < 0.05$ (uncorrected).

Metric	MG	G3	Δ	p	r
Precision	0.370	**0.400**	−0.029	<0.001*	0.805
Recall	0.692	**0.811**	−0.119	<0.001*	0.972
F1	0.453	**0.509**	−0.056	<0.001*	0.828

Table 3: Sub-task 3: Admission Summary Generation ($n = 637$ pairs). Filtered reference = hospital course sentences with at least one term in `source_facts`. Bold = winning model per row; $* = p < 0.05$ (uncorrected). Mention-rate metrics measure named-substring presence, not numerical fidelity.

Metric	MG	G3	Δ	p	r
ROUGE-L	**0.110**	0.101	+0.009	<0.001*	0.743
BERTScore F1	**0.739**	0.732	+0.007	<0.001*	0.808
Problem Coverage	0.068	**0.069**	−0.001	0.672	0.935
Out-of-Range Mention Rate	0.235	**0.314**	−0.079	<0.001*	0.964
Medication Mention Rate	0.452	**0.481**	−0.029	0.003*	0.800

4.3 Sub-task 3: Admission Summary Generation

Table 3 reports all five Sub-task 3 metrics; Figure 3 shows per-record distributions. The pattern reverses partially: MedGemma-4B-IT achieves higher ROUGE-L and BERTScore F1, consistent with its reported pre-training on clinical narrative text [3]. However, Gemma-3-4B-IT names a higher fraction of out-of-range metrics and administered medications: Out-of-Range Term Mention Rate ($\Delta = -0.079$, $p<0.001$) and Medication Name Mention Rate ($\Delta = -0.029$, $p=0.003$). Problem Coverage is near-floor for both models (≈7%) with no significant difference ($p = 0.672$), reflecting the difficulty of verbatim ICD-9 description matching. We caution that the mention-rate metrics reward vocabulary overlap, not numerical fidelity or clinical correctness; the same verbosity-inflates-mentions mechanism noted for Sub-task 2 likely contributes here.

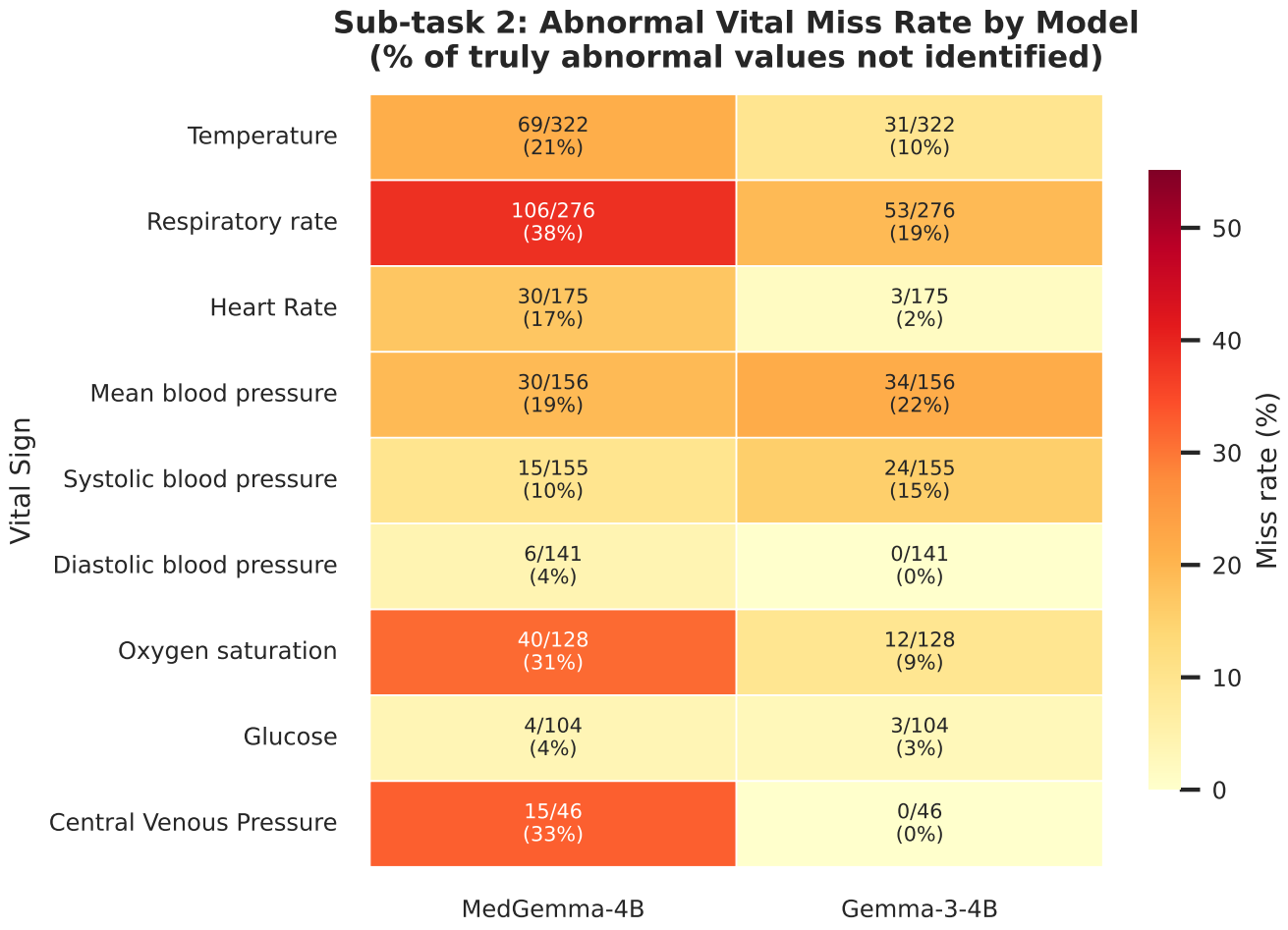

Figure 2: Per-vital miss rates for Sub-task 2. Darker cells indicate higher miss rates. MedGemma-4B-IT misses Heart Rate (17%), Respiratory Rate (38%), and Oxygen Saturation (31%) at substantially higher rates than Gemma-3-4B-IT (2%, 19%, and 9% respectively). MAP miss rates are comparable between models (19% MedGemma, 22% Gemma-3).

4.4 Cross-task Summary

Of 13 evaluated metrics across three sub-tasks, we classify two Sub-task 1 results (ROUGE-L, aggregate Entity F1) as draws despite reaching nominal significance, on the grounds that the absolute deltas (+0.003 and +0.005) are below practical-meaning thresholds and the metrics are near-floor or recogniser-limited. MedGemma-4B-IT retains a significant advantage on 2 metrics (ROUGE-L and BERTScore F1 in Sub-task 3, both stylistic- similarity measures with modest absolute gains). Gemma-3-4B-IT retains a significant advantage on 7 metrics (Entity F1 vitals and AARP Score in Sub-task 1; Precision, Recall, and F1 in Sub-task 2; Out-of-Range Term Mention Rate and Medication Name Mention Rate in Sub-task 3). The remaining 4 metrics are draws or not significantly different (Sub-task 1 ROUGE-L and aggregate Entity F1 reclassified as draws; Sub-task 1 Drug Entity F1 and Sub-task 3 Problem Coverage not significant). The AARP and mention-rate results carry the caveats discussed in the preceding sections.

5 Discussion

5.1 Null-to-Negative Result Interpretation

This study is an explicit null-to-negative result, and we consider that its most important contribution. The common inference pattern, "this model was pre-trained on clinical data, therefore it should perform better on clinical tasks," is not supported by our data. At the 4B parameter scale, with identical architecture and inference parameters, the general instruction-tuned model matches or exceeds the clinically pre-trained model on the majority of clinically meaningful metrics.

We hypothesise two explanations. First, the tasks we evaluate (structured output compliance, AARP format, precise numerical grounding, and instruction following) are primarily *capability* requirements, not *knowledge* requirements. A model that generates a correctly structured AARP block with accurate vital sign values succeeds through instruction adherence and pattern recognition, not through medical knowledge retrieval. The clinical pre-training

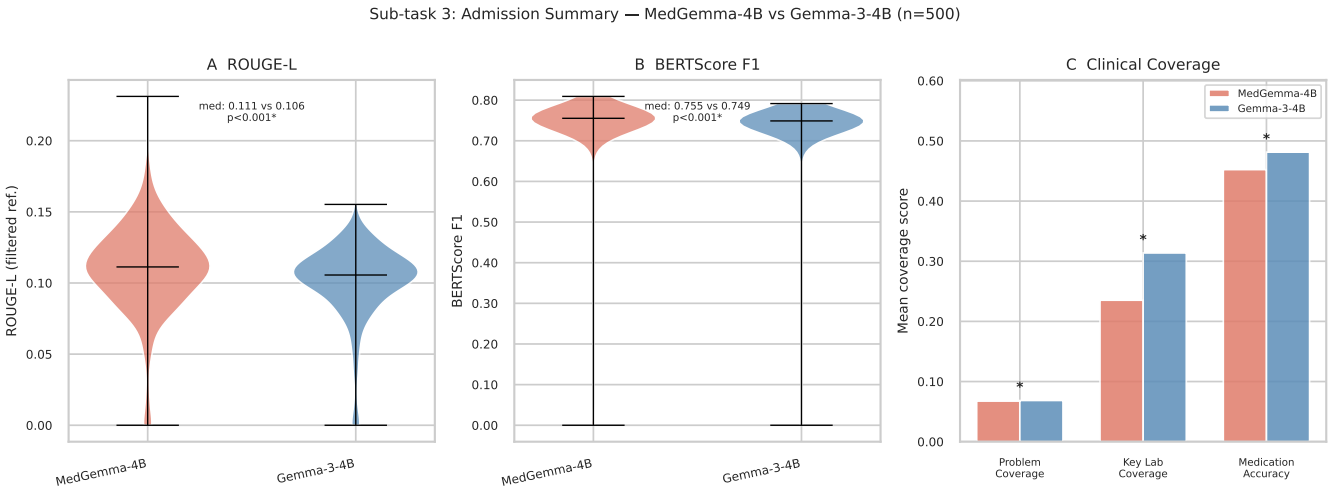

Figure 3: Sub-task 3 results. (A) ROUGE-L violin plots. (B) BERTScore F1 violin plots. (C) Clinical mention-rate bar chart. MedGemma-4B-IT achieves marginally higher lexical/semantic similarity; Gemma-3-4B-IT substantially outperforms on out-of-range term mention rate. $* = p < 0.05$.

corpus likely contained extensive unstructured clinical narratives, which may have conditioned MedGemma-4B-IT away from rigid template adherence.

Second, the MedGemma-4B-IT repetition pathology (looping under long structured prompts) suggests that the clinical pre-training or alignment process may have introduced a sensitivity to long, repetitive structured contexts that the general model does not exhibit. This is a practical limitation: a model that requires a repetition penalty to produce any coherent output is operationally less reliable than one that does not.

5.2 Sub-task 3 Mixed Pattern

Sub-task 3 is the only task where MedGemma-4B-IT shows a significant advantage, and the advantage is on *stylistic* metrics (ROUGE-L, BERTScore), not on named mention of input facts. This is consistent with the explanation above: MedGemma-4B-IT's pre-training corpus reportedly includes clinical notes and discharge summaries [3], which share the prose style of the hospital course reference text. The model has learned to generate discharge-summary-like prose, but that prose names fewer of the specific out-of-range metrics and medications from the structured input than Gemma-3-4B-IT's output does. Because our mention-rate metrics measure substring presence rather than numerical fidelity, we frame this as a difference in *vocabulary alignment with the input* rather than as clinical grounding; nevertheless, in a clinical safety context, naming the relevant out-of-range metric is a necessary (if not sufficient) condition for grounded documentation.

5.3 Limitations

Independence by admission. This evaluation samples one record per unique hospital admission ($n = 637$ unique admissions). This design satisfies the paired independence assumption of the Wilcoxon test and eliminates the within-admission ICD-9 correlation present in note-level sampling. Results represent 637 statistically independent patients.

ROUGE-L and Entity F1 floor effect. Sub-task 1 ROUGE-L is 0.076–0.079 for both models, near-floor; aggregate Entity F1 is similarly low at 0.054–0.059. The ground-truth nursing notes are written from direct bedside observation (physical examination, subjective patient reports) that is entirely absent from the structured EHR context. No model can recover this information. Neither metric is a valid quality signal for this task; we retain them only for comparability with prior work and reclassify them as draws.

Inference configuration confound. We applied an identical repetition penalty of 1.15 to both models because preliminary diagnostics showed MedGemma-4B-IT degenerated into truncated loops without it. We did not subsequently sweep the penalty per model. Each model's performance ceiling at its own optimal inference configuration is therefore unknown, and the AARP Score result in Sub-task 1 in particular should be read as "MedGemma-4B-IT fails AARP at the only configuration we found that produced any non-truncated output for it," not as a configuration- independent capability claim.

Sub-task 2 verbosity confound. The Sub-task 2 metric counts a vital sign as identified if its label appears in the generated text, without requiring the model to flag the value as abnormal. A more verbose model can therefore inflate recall by naming more metrics in preamble. We did not re-score with a stricter explicit-flagging rule.

Sub-task 3 mention-rate metrics. Out-of-Range Term Mention Rate and Medication Name Mention Rate measure substring presence of metric labels and medication first-tokens respectively. They do not check that numeric values are accurate, that medications are described as administered, or that out-of-range values are explicitly identified as abnormal. They reward vocabulary alignment with the input, not clinical grounding in a strong sense.

BERTScore encoder. BERTScore F1 was computed with non-cased distilbert-base, a general-domain distilled encoder that can over-reward stylistic fluency relative to clinical specificity. A clinical encoder (e.g., `Bio_ClinicalBERT`) may yield different relative scores between the two models on Sub-task 3.

Single ICU type and institution. All records are from the MICU at a single Boston academic medical centre. Nursing note conventions, vital sign reference ranges, and medication administration practices differ across ICU types (cardiac, surgical, neonatal) and institutions. Generalisability is unknown.

Parameter scale. Both models are 4B parameters; this is simultaneously a design strength and a limitation. The controlled comparison (identical size, identical architecture, identical inference parameters) isolates pre-training corpus as the single experimental variable, but the results speak only to this scale. Two distinct scale effects remain unexplored: (1) whether the *absolute capability floor* at 4B suppresses task performance for both models equally, masking pre-training effects that emerge only at higher capacity; and (2) whether the *relative ordering* between clinical and general pre-training is stable across scales. We observed that MedGemma-4B-IT's primary failure mode (AARP instruction non-compliance) is a general instruction-following deficit; this class of failure tends to diminish at larger parameter counts for all models. However, Gemma-3-4B-IT's larger variants (12B, 27B) also benefit from the same scaling, so the relative advantage is not guaranteed to reverse. Evaluating MedGemma-4B-IT-27B against Gemma-3-4B-IT-27B would be the natural follow-up.

Single checkpoint pair. We compare exactly two checkpoints. They differ in pre-training corpus, but also in alignment recipe, instruction-tuning data, RLHF, and possibly base model snapshot. Our design therefore falsifies the vendor's specialization claim for this checkpoint pair without isolating the pre-training corpus as the causal driver. Generalisation to other clinical-vs.-general pairs (e.g., Meditron-7B vs. Llama-2-7B; BioMistral vs. Mistral-7B) is left for future work.

No multiple-comparisons correction. We report uncorrected p-values across 13 metrics. Several of the marginal results in Sub-task 1 (ROUGE-L, aggregate Entity F1) have absolute deltas below practical-meaning thresholds; we have reclassified these as draws in the cross-task summary, but readers applying a Holm or BH-FDR correction should not be surprised if some borderline significances do not survive adjustment.

6 Conclusion

We evaluated MedGemma-4B-IT against Gemma-3-4B-IT on three MICU nursing tasks from MIMIC-III, using one record per unique hospital admission ($n = 637$ independent records) and 13 metrics. For this checkpoint pair at the 4B parameter scale, the clinically marketed model does not provide a consistent performance advantage. After reclassifying near-floor Sub-task 1 metrics as draws, Gemma-3-4B-IT significantly outperforms MedGemma-4B-IT on 7 of 13 metrics, including AARP document compliance (subject to an inference-configuration caveat), abnormal value recall and precision, and named mention of out-of-range metrics and medications in summaries. MedGemma-4B-IT retains significant but modest advantages on stylistic similarity in free-form prose (ROUGE-L and BERTScore F1 in Sub-task 3). Because our design varies both pre-training corpus and alignment recipe between the two checkpoints, we cannot attribute this pattern to pre-training corpus alone.

These findings have a practical implication: for resource-constrained healthcare settings evaluating this specific 4B-parameter checkpoint pair for MICU documentation support, the evidence does not justify a preference for MedGemma-4B-IT on the basis of vendor specialization claims. We hypothesise, without proving, that the more important variable is instruction-following quality, which is shaped by alignment training rather than pre-training corpus composition.

Future work. Two extensions would strengthen these findings: (1) evaluate MedGemma-4B-IT-27B against Gemma-3-4B-IT-27B to test whether the relative ordering is stable across parameter scales; (2) conduct a prospective human evaluation in which ICU nurses rate clinical accuracy and usability of generated notes, which would provide the ground truth that automated metrics cannot. We also identify the automated detection of ICD-9 problem coverage as an open metric design problem: verbatim description matching is too strict; clinical concept matching via UMLS normalisation would be more informative.

Reproducibility

Code, evaluation scripts, and figure generation are available at https://github.com/lauesa/ai-healthcare-52680. MIMIC-III data requires a credentialed PhysioNet account and cannot be redistributed.

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A Prompt Templates

All prompts follow the chat format: a fixed system message sets the model persona and constraints; a user message provides the structured patient context assembled from the FHIR-lite record (Section 3). Placeholders in angle brackets represent fields populated at runtime.

Sub-task 1: Nursing Note Generation

System:

> You are an experienced MICU nurse writing an end-of-shift progress note. Use clinical shorthand. Do not invent medications, procedures, or diagnoses not present in the context. Do not include a document header.

User:

> *[Admission context: patient identifier, care unit, admitting diagnosis, admit time offset]*
> *[Active problems: ranked ICD-9 diagnosis list]*
> *[Observations: vital signs from the 12-hour window, with out-of-range flags, capped at 1,100 entries]*
> *[Labs: laboratory values from the 12-hour window, capped at 150 entries]*
> *[Medications: CareVue administration events, capped at 250 entries]*
> *[Procedures: CareVue procedure events]*
>
> **Example note (for format reference only; do not copy content)**
>
> Impaired Skin Integrity
> Assessment:
> All skin surfaces as documented in flowsheets. Some sloughing on buttocks, pt stooling loose bilious green BM.
> Action:
> Buttocks cleansed, antifungal cream and adaptic applied. Flexiseal ordered and inserted.
> Response:
> Buttock areas free of fecal contamination overnoc, areas appear slightly less irritated.
> Plan:
> Cleanse and treat impaired skin areas, protect open areas from friction and fecal contamination.
>
> Renal failure, acute (ARF)
> Assessment:
> CRRT as ordered, no difficulties with circuit. UO via foley 10–15 cc clear yellow.
> Action:
> Removed 1–200 cc/h. Ca and K gtts titrated per labs.
> Response:
> Attained fluid removal goal, neg 2700 cc yesterday. Hemodynamically stable.
> Plan:
> Fluid removal goal 2.5–3 L/day. Monitor lytes q6 and titrate repletion gtts as ordered.
>
> **Task**
> Write the nursing progress note. One AARP block per active problem. Synthesise the data; do not restate it verbatim.

The few-shot example is drawn from a de-identified MIMIC-III nursing note in the validation split and is included in the user turn so the model treats it as a concrete format demonstration.

Sub-task 2: Abnormal Value Identification

System:

> You are an experienced MICU nurse reviewing a patient's vital signs. Identify which values are clinically abnormal for an ICU adult patient. For each abnormal value: (1) state whether it is high or low, (2) give the normal adult range with units, (3) name the clinical concern in one sentence. If a value is within normal limits, do not mention it. Be concise. Use clinical shorthand.

User:

> **Patient Context**
> Admitting diagnosis: *[admit diagnosis]*
> Care unit: *[care unit]*
>
> **Most Recent Vital Signs**
> *[One line per metric: name, value, unit, time offset]*
>
> **Task**
> Identify and explain any clinically abnormal vital signs above.

Only the nine metrics with hardcoded ICU reference ranges are included (Heart Rate 60–100 bpm; Systolic BP 90–140 mmHg; Diastolic BP 50–90 mmHg; MAP 65–105 mmHg; Respiratory Rate 12–20 breaths/min; $SpO_2 \geq 95\%$; Temperature 97.7–99.5 °F; CVP 2–12 mmHg; Glucose 70–180 mg/dL). These same ranges define the ground-truth labels used for evaluation and are independent of the statistical outlier bounds applied during ETL.

Sub-task 3: Admission Summary Generation

System:

You are an experienced MICU attending physician writing the Hospital Course section of a discharge summary. Synthesise the structured ICU data provided into a concise problem-oriented narrative. For each active problem: describe the clinical findings, key investigations, interventions, and patient response. Use clinical language. Be specific: cite abnormal values, medications, and procedures by name. Do not use bullet points or AARP headings; write flowing prose.

User:

[Admission context, active problems, observations, labs, medications, procedures; identical fields to Sub-task 1]

Task

Write the Hospital Course narrative for this 12-hour ICU window. One paragraph per active problem. Synthesise the data; do not restate it verbatim.